BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 278 · SQL SELECT, JOIN and nested SELECT

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is the purpose of an INNER JOIN?

Q2. Which relational key connects two tables?

Q3. What does WHERE do?

Q4. Binary search for 9 in sorted [1,3,5,7,9,11,13]. Show midpoint indexes.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

SELECT — Choose fields for output.

INNER JOIN — Combine matching rows.

nested SELECT — Query inside another query.

IN — Compare value against returned set.

WHERE — Filter rows.

New learning

Understand — SELECT names output columns, FROM identifies input tables and WHERE restricts rows using a Boolean predicate.

Trace — INNER JOIN combines rows matching a join condition. A nested SELECT can return a set of keys used by an outer IN condition.

Explain — Both example queries return Ada and Ben, but joining is more flexible when fields from both tables are required.

Worked example

Write SQL for student names where Score.Mark>80.

1. Predict — Only Ben qualifies because score 82 is the sole result above 80.

2. Check the answer — Explain which rule makes this result correct.

We Do · Practise with guidance

Write a nested SELECT for student names scoring at least 75.

```
SELECT Name FROM Student WHERE ID IN
(SELECT ID FROM Score WHERE Mark>=75);
```

Predict

Subquery returns IDs 1 and 2

Check the answer

outer query returns Ada and Ben.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Write SELECT to show all Name values in Student sorted by Name ascending. [4]

Question 2

Write and trace a join returning Name and Mark for scores >=70. [6]

Question 3

Write a nested SELECT for students scoring above 75. Explain result. [6]

Exit Ticket · Answer before checking

INNER JOIN results can be the same as a nested key SELECT for a particular data set.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Combines matching rows.

Q2. Foreign key referencing primary key.

Q3. Filters results according to a predicate.

Q4. low0 high6 mid3→7 then low4 high6 mid5→11 then high4 mid4→9, found index4.

You Do pages · Independent questions

Q1. SELECT Name [1] FROM Student [1] ORDER BY Name (ascending) [2]; explicitly writing ASC is optional because SQL defaults to ascending order.

Q2. SELECT Student.Name,Score.Mark [1] FROM Student [1] INNER JOIN Score [1] ON Student.ID=Score.ID [1] WHERE Score.Mark>=70 [1]; output Ada75, Ben82 [1].

Q3. SELECT Name FROM Student WHERE ID IN (SELECT ID FROM Score WHERE Mark>75) [4]; inner query returns only ID2 [1], outer returns Ben [1].

Exit Ticket · Model answer

True. Different query forms can yield equivalent selected results.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.